Digital Stubs

Software Requirements Specification

4.0

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Group 4
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Revision History

Date	Description	Author	Comments
2/19/2026	Version 1.0	Jasper McIntyre	Requirements Specification first draft
3/5/2026	Version 2.0	Jasper McIntyre	Added SDS
3/19/2026	Version 3.0	Jasper McIntyre	Added test plan
4/9/2026	Version 4.0	Jasper McIntyre	Added SWA diagram description and Data Management Strategy

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
	Jasper McIntyre	Software Eng.	
	Dr. Gus Hanna	Instructor, CS 250	

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1. Introduction

1.1 Purpose

The purpose of this software requirement specification is to give a complete description of every feature and constraint for the Digital Stubs program. The target audience is software engineers and movie theater management.

1.2 Scope

Digital Stubs is a digital ticketing system made to replace physical movie tickets with secure and trackable digital tickets. Customers can find when movies are showing, choose their seats, purchase tickets, and store them in a secure digital token. This platform will not handle physical tickets or concessions. The goal with this project is to prevent lost tickets, reduce waste, and increase the rate that customers can be checked in.

1.3 Definitions, Acronyms, and Abbreviations

Digital Stub: The digital representation of a paper movie ticket that incorporates a QR code

QR Code: Quick Response Code; Two dimensional barcode used as a secure digital identifier

AES: Advanced Encryption Standard; A symmetric-key cipher used by the ticket program to encrypt and protect digital stubs

1.4 References

Advanced Encryption Standard: National Institute of Standards and Technology (NIST) Federal Information Processing Standards Publication 197

CS 250 Notes: Use Case Outline and Software Requirement Template (2026)

IEEE Standard 830-1998: IEEE Recommended Practice for Software Requirements Specifications (1998)

1.5 Overview

The rest of this SRS has 4 sections:

2) A detailed description of the Digital Stubs program and how it is used
3) The functional and non functional requirements for usage and testing.
4) Analysis models like Use Case Diagrams
5) The process used to apply future changes

2. General Description

2.1 Product Perspective

Digital Stubs is a standalone program designed to work with third party mobile wallet APIs like Apple Pay and Google Wallet

2.2 Product Functions

Handle user registration and authentication

Provide real time movie times with a seat selection interface

Encrypt individual ticket information

Employee UI for ticket scanning and tracking

2.3 User Characteristics

Movie customers who prefer digital formats to paper tickets that can be lost

Theater staff have a quick and simple to use system for checking tickets

Managers will be able to track and sort sales data much easier

2.4 General Constraints

The system requires QR code scanners that work in low light as well as compatible devices to host the program and data on

2.5 Assumptions and Dependencies

The program assumes that users have access to a modern smartphone with capable wallet capabilities. The system relies on the constant availability of the theater's seat selection userbase and external payment programs.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

3.1.2 Hardware Interfaces

3.1.3 Software Interfaces

3.1.4 Communications Interfaces

3.2 Functional Requirements

3.2.1 Interactive Seating Selection

3.2.1.1 Introduction: Displays visual of seat choices for users
3.2.1.2 Inputs: User selection and selected movie
3.2.1.3 Processing: Database for seat availability displayed with green and red seats
3.2.1.4 Outputs: Digital map of the auditorium
3.2.1.5 Error Handling: Denying access to selected seats

3.2.2 Digital Stub Creation

3.2.2.1 Introduction: Creates scannable ticket
3.2.2.2 Inputs: Transaction record, seat number, User ID
3.2.2.3 Processing: Data encryption to user data and store as QR code
3.2.2.4 Outputs: Digital Stub token sent to user email and e-wallet

3.3 Use Cases

3.3.1 Use Case #1: Ticket Purchase

When a user logs in and chooses a showtime and seat, the data flows from the seat selection, payment, data encryption, and QR code sending.

3.3.2 Use Case #2

When a theater staff logs a users ticket, the data flows from the QR code, the system decrypts it, marks their seat as chosen, and plays the employee a positive noise.

3.4 Classes / Objects

3.4.1.1 Attributes
- Stub ID: a unique identifier for the users ticket
- Seat Selection: an identifier representing the seat number
- User Hash: An AES encrypted string used to make the QR code
- Seat Status: A boolean value representing whether a seat is open or taken
-

3.4.1.2 Functions
- Generate QR: converts the User Hash to a QR code
- User Entry: checks the Seat Status and system time to grant the user ticket access
-

3.5 Non-Functional Requirements

3.5.1 Performance: User authorization and ticket validation happen in seconds

3.5.3 Availability: Requires a constant wifi connection and up to date, compatible devices

3.5.4 Security: All stubs must be AES encrypted to stop theft or duplicates

3.5.5 Maintainability: Must be able to adapt to new forms of NFC technology

3.6 Inverse Requirements

The system will never store raw user data or payment information

3.7 Design Constraints

The system must use AES encryption standards to secure and protect user data

3.8 Logical Database Requirements

The system will use a Relational Database Management System to store movie schedules and seat selections

3.9 Other Requirements

The user interface will work will multiple languages and different phone, wallet types

4. Analysis Models

4.1 Sequence Diagrams

- The diagram shows the process of user movie selection, seat choice, and payment

4.3 Data Flow Diagrams (DFD)

4.2 State-Transition Diagrams (STD)

5. Change Management Process

Any Digital Stub developer may submit change tickets within themselves to make changes

A. Appendices

A.1 Appendix 1

Includes drafts of the user seat map and the employee ticket portal

A.2 Appendix 2

Group brainstorm where we initially decided to replace paper tickets with a digital system

6. Software Architecture Overview

6.1 System Description

The Digital Stubs system is organized into a 3 tier architecture that separates the user interface from data and logic which makes sure that developers can repair different components without having to shut down the whole system. The system allows customers to browse showtimes and select seats, complete payments, and receive a QR code representation for their digital stub. Theater staff use a portal to scan and validate tickets at the entrance. Theater managers can easily access sales and attendance data.

6.2 Software Architectural Overview

The system follows a 3 tier architectural model that ensures a clear separation between user interaction from the core logic and data storage.

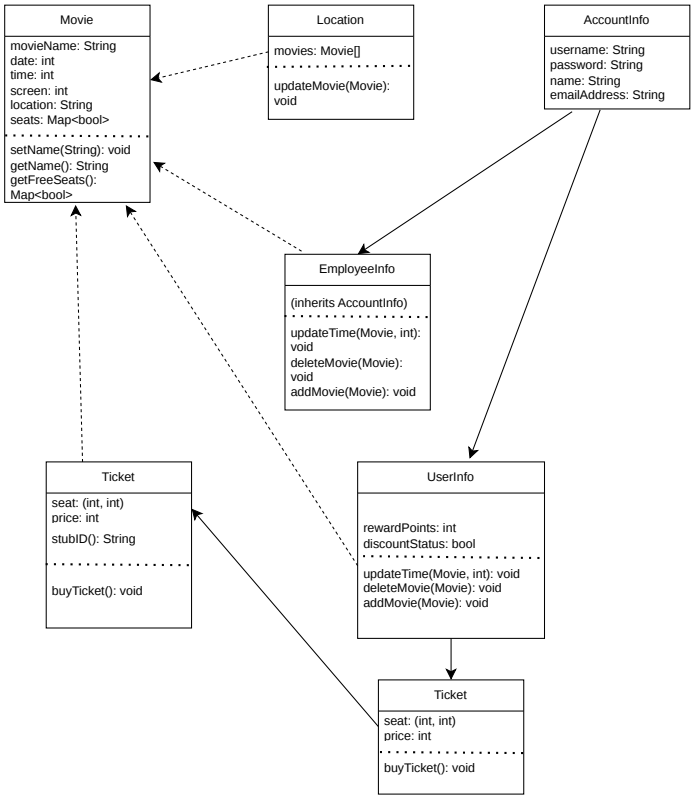
<u>Presentation Layer</u>: Contains the customer mobile interface that integrates with Apple Wallet and Google Wallet APIs, and the employee UI used for scanning and validating tickets at the theater entrance.

Application Layer: The central logic component that handles AES encryption to generate secure QR codes, manages real time seat availability, processes payments, and communicates between every layer.

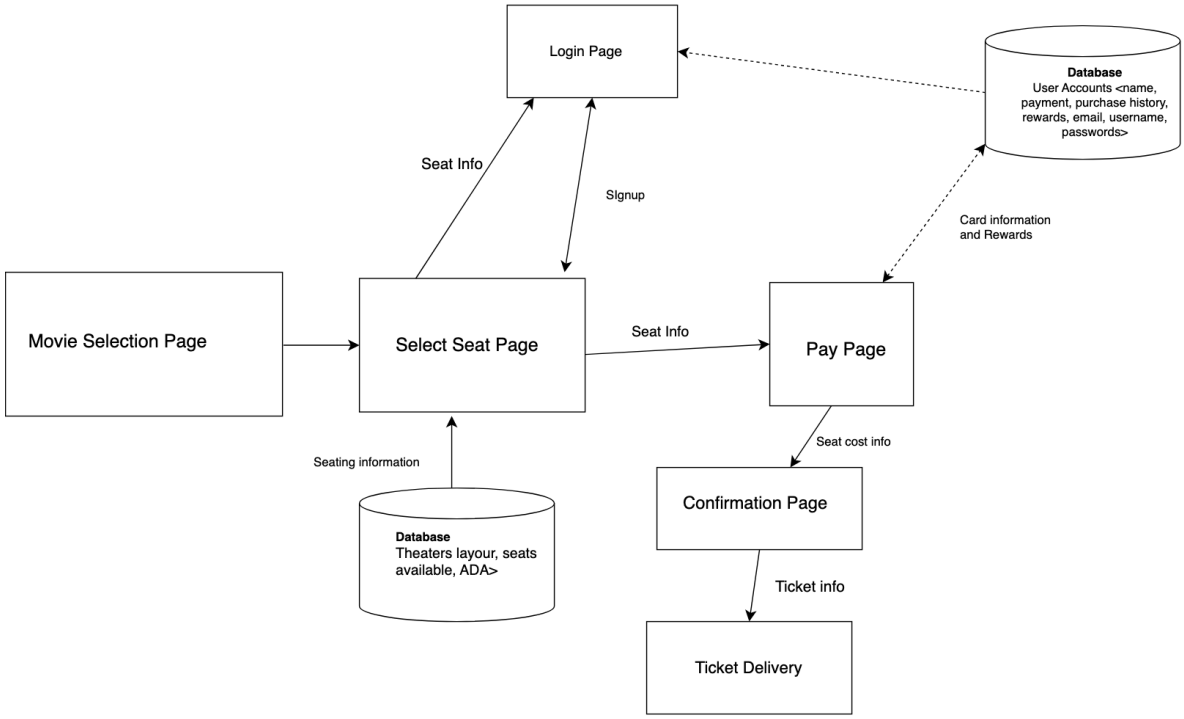
Data Layer: A relational database that stores movie schedules, auditorium layouts, user records, and transaction history.

Connectors and Protocols:
- HTTPS/TLS: Secures data transmission between the application server and the user interface
- QR/NFC: The communication method used between the user's digital wallet and the theater's scanning hardware
- JDBC: Allows the system's logic to communicate with the database to read movie showtimes and update seat availability

6.3 UML Class Diagram



6.4 SWA Diagram



6.5 Development Plan and Timeline

Phase	Task	Owner	Duration
1	Database setup	Jasper	1 week
2	User authentication and login	Jasper	1 week
3	Showtime and seat selection	Jasper	1 week
4	AES encryption and QR code generation	Jasper	1 week
5	Payment processing	Jasper	1 week
6	Customer UI	Jasper	1 week
7	Staff scanner portal	Jasper	1 week
8	Integration testing	Jasper	1 week

7. Test Plan

https://github.com/jmcintyre568/250update

7.1 Purpose and Scope

The goal is to verify that the Digital Stubs system meets the technical specifications for verification and that the customer can enter the theater with a secure token using validation

Verification: We will use Unit Testing to verify that the AES encryption logic correctly transforms user data into secure hashes without loss

Validation: We will use System Testing to validate the end to end user experience from the seat selection page to to theater entry

7.2 Target Features and Features to be Tested

Target: AES encryption logic, real time seat selection, 3 tier communication from presentation to data layer

Test Vectors:
Normal: Valid payments and available seats
Boundary: Booking the last available seat in an auditorium
Illegal: Trying to scan a QR code with an expired Stub ID

7.2.1 Failure Coverage

Testing will focus on preventing double bookings with race conditions in the data layer and ticket cloning by ensuring the AES encryption prevents duplicates

7.3 Updated SWA Diagram Description

The SWA diagram maintains the 3 tier architectural model.

Presentation Layer: Movie Selection Page; Also includes the Select Seat Page, Login Page, Pay Page

Application Layer: Handles logical routing for seat information and signups. Also manages seating information and seat cost data. It also handles ticket delivery.

Data Layer: Visualizes separate logically split database nodes

Theater DB: Receives queries from the Select Seat Page. Also manages theater layouts, available seats, and ADA data

User DB: Receives queries from the Login Page and Pay Page. It manages user accounts, names, purchase history, rewards, usernames and passwords.

7.4 Data Management Strategy

7.4.1 Database Type: SQL vs Non SQL

The Digital Stubs system will utilize a Relational Database Management System using SQL for its core data layer. SQL is chosen to guarantee strict ACID compliance and generate secure AES encrypted digital stubs.

7.4.2 Design Decisions

Based on the architecture, the data layer is split into two databases

Theater and Scheduling database: Manages theater layouts. It also manages real time seat availability and ADA compliance information

User Database: Stores user accounts, passwords, purchase history, and rewards

During the design phase, we also considered a Single Monolithic SQL Database that would combine all tables into a single database instance.

We chose a monolithic database because it's simpler to deploy and maintain. However, splitting the data into a Theater and User database allows the system to scale seat selection queries separately from user authentication and payments. This reduces the risk of bottlenecking during major movie releases.

While using No SQL could offer lower latency for seat selection, using only SQL based RDBMS was chosen to ensure data integrity.